Aware Lecture Summarization and Flashcard Generation for Intelligent Study Assistants

R26-SE-022

Project Proposal Report

Vijithapala T.G.K.G – IT22187178

Supervised By Mrs. Suriya Kumari

Co-supervised By Mr. Ravi Supunya

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Software Engineering

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Vijithapala T.G.K.G	**IT22187178**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Mrs. Suriya Kumari	2026/03/13	

ABSTRACT

University students often struggle to revise large volumes of lecture materials during examination periods, as slides, notes, and supplementary readings often contain extensive information that makes it difficult to identify key concepts and exam-relevant topics efficiently. Although artificial intelligence (AI)–based summarization tools have been introduced to assist learners, many existing solutions produce generic summaries that do not consider course learning objectives or assessment requirements, resulting in the omission of important examinable concepts and the inclusion of less relevant information. To address this limitation, this research proposes a learning-objective-aware lecture summarization and flashcard generation system designed to support efficient academic revision. The system will analyze lecture slides, transcripts, and course syllabi to identify critical concepts and align generated summaries with predefined learning objectives. Transformer-based Natural Language Processing (NLP) models will be employed to produce concise summaries, which will then be converted into flashcards to promote active recall and improve knowledge retention. The proposed approach is expected to enhance revision efficiency, reduce information overload, and improve learning outcomes for university students, with potential applications in educational technology platforms and intelligent tutoring systems that provide automated study assistance.

Keywords: AI summarization, educational NLP, flashcard generation, intelligent study assistant, learning-objective-aware summarization

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Mrs. Suriya Kumari, for her continuous guidance, valuable feedback, and encouragement throughout the development of this research proposal. Her insights and expertise have been instrumental in shaping the direction of my individual component.

I also extend my appreciation to my co-supervisor, Mr. Ravi Supunya, for his constructive suggestions and support during the early stages of this research.

My heartfelt thanks go to my project group members for their collaboration, discussions, and shared commitment toward achieving the overall objectives of our research project.

Finally, I am grateful to the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment, resources, and learning opportunities that enabled me to undertake this research.

TABLE OF CONTENT

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 7

LIST OF FIGURES 8

LIST OF ABBREVIATIONS 9

LIST OF APPENDICES 10

1.INTRODUCTION 11

1.1 Background and Context 11

1.2 Literature Review 14

1.3 Research Gap 16

2.1 OBJECTIVE 18

2.1 Main Objective 18

2.2 Specific Objectives 18

3. METHODOLOGY 19

3.1 Research Approach 19

3.2 Technical Methods and Algorithms 19

3.2.1 Data Preprocessing 19

3.2.2 Lecture Summarization 19

3.2.3 Learning Objective Alignment 20

3.2.4 Flashcard Generation 20

3.3 Tools and Platforms 20

3.4 Data Collection and Ethical Considerations 21

3.5 Validation Strategy and Performance Metrics 21

4. HIGH-LEVEL SYSTEM ARCHITECTURE 22

4.1 Overview of Proposed Solution 23

4.2 Component Description 24

4.3 Component Interactions 25

4.4 Data Flow Explanation 25

4.5 Scalability and Security Considerations 25

5. USER REQUIREMENTS 26
5.1 Stakeholder Identification 26
5.2 Functional Requirements 26
5.3 Non-Functional Requirements 27
6. COMMERCIALIZATION PLAN 29
6.1 Target Market and Customer Persona 29
6.2 Value Proposition 29
6.3 Revenue Model 30
6.4 Cost Estimation 30
6.5 Pricing Strategy 30
6.6 Competitive Advantage 31
6.7 Intellectual Property Considerations 31
7. BUDGET AND JUSTIFICATION 32
8.WORK BREAKDOWN STRUCTURE (WBS) 34
9.GANTT CHART 35
10. References List 36
9. APPENDICES 37
Appendix A – Survey Instrument 37
Appendix B – System Diagrams 37
Appendix C – Preliminary UI Mockups 38
Appendix D – Risk Analysis 38
Appendix E – Ethical Considerations 38

LIST OF TABLES

1 Hardware Costs 32
2 Software Licenses 33
3 Cloud Services 33

LIST OF FIGURES

Figure 4. 1 **High-Level Architecture of the Proposed Lecture Summarization System** ..23
Figure 4. 2 **Internal Architecture of the Lecture Summarization and Flashcard Generation Module** 24

LIST OF ABBREVIATIONS

Abbreviation	Full Form
S3	Amazon Simple Storage Service
SDG	Sustainable Development Goal
SaaS	Software as a Service
LLM	Large Language Model
NLP	Natural Language Processing
GPU	Graphics Processing Unit

LIST OF APPENDICES

Appendix	Title	Page
Appendix A	Survey Instrument	37
Appendix B	System Diagrams	37
Appendix C	Preliminary UI Mockups	37
Appendix D	Risk Analysis	37
Appendix E	Ethical Considerations	37

1.INTRODUCTION

1.1 Background and Context

- **Real-World Problem & Who is Affected**

  Students in higher education institutions worldwide face significant challenges when reviewing large volumes of academic materials in preparation for examinations. Modern university environments generate substantial content across multiple formats, including lecture slides, transcripts, recorded sessions, and supplementary reading materials. For undergraduate students in particular, the volume of content produced each semester makes it increasingly difficult to identify key concepts, prioritize exam-relevant topics, and organize effective revision strategies within limited time. This problem leads to information overload, poor time management during examination periods, and ultimately lower academic performance among students who lack structured tools to guide their revision process.

- **Statistical Evidence**

  The scale of this problem is reflected in global and regional data on student academic performance and learning behavior. According to a UNESCO report, approximately 617 million children and young people worldwide lack basic learning proficiency, highlighting the urgent need for improved learning support mechanisms at all levels of education. Studies in higher education contexts further reveal that students spend an average of 15 to 20 hours per week reviewing academic materials yet report low confidence in their ability to identify the most important content before examinations. In the Sri Lankan higher education context, institutions such as SLIIT accommodate thousands of undergraduate students across technology disciplines, where the volume of lecture content per module can exceed hundreds of slides per semester, placing significant revision burdens on students without adequate AI-assisted tools to support them.

- ## Analysis of Existing Solutions and Their Limitations

Recent advances in Artificial Intelligence (AI), particularly in Natural Language Processing (NLP), have enabled automated systems capable of extracting and summarizing important information from large volumes of textual content [1]. These systems have been applied in academic settings to help students process educational materials more efficiently. AI-powered student learning systems have also been developed to support personalized learning by analyzing course content and recommending relevant study resources [2]. Furthermore, the integration of AI and NLP within e-learning platforms has enabled automated content delivery, intelligent feedback mechanisms, and interactive learning support within digital education environments [3].

However, a fundamental limitation persists across most existing AI-based summarization tools. General-purpose systems such as ChatGPT and other widely used summarization software condense text based on linguistic patterns and statistical frequency rather than pedagogical relevance. This means their output often fails to reflect what is actually important within a specific academic course. Generative AI tools have demonstrated strong potential in educational content generation, yet researchers consistently highlight that their outputs lack alignment with curriculum-level learning outcomes [4]. AI-powered study companion systems can generate summaries and flashcards from course content, but these materials are typically created without reference to exam requirements or course objectives [5]. Intelligent tutoring systems offer personalized explanations and adaptive support, but similarly do not produce structured revision materials mapped to expected learning outcomes [6]. More advanced retrieval-augmented generation models improve the accuracy of AI responses by combining information retrieval with language generation, yet remain focused on tutoring interactions rather than structured summary generation [7]. AI-enabled learning management systems that generate quizzes and flashcards dynamically still rely on general content processing rather than objective-aligned prioritization [8]. Even conversational LLM-driven educational chatbots, such as Moodle Bot, which assist students with course-related questions, do not specifically generate summaries or flashcards aligned with defined learning objectives [9]. As a result, students using these existing tools risk spending revision time on less critical content while overlooking the key examinable concepts required for effective academic preparation.

- ## Proposed Solution and Its Parts

To address this limitation, the proposed research introduces a Learning-Objective-Aware Lecture Summarization and Flashcard Generation System. The system accepts lecture slides, transcripts, and course syllabi as inputs and processes them through a pipeline of interconnected modules. A document processing module extracts textual content from uploaded academic materials. An NLP preprocessing module performs tokenization, stop-word removal, and sentence segmentation to prepare the text for analysis. A transformer-based summarization engine, using models such as BART or T5, then generates concise summaries of the lecture content. A learning objective alignment module applies semantic similarity techniques and embedding models to compare lecture sentences against predefined course learning objectives, ensuring that the most exam-relevant concepts are prioritized in the output. Finally, a flashcard generation module converts the aligned summarized content into question-and-answer pairs to support active recall learning, which is a well-established technique for improving long-term knowledge retention. Together, these components form an end-to-end intelligent revision system tailored specifically to the academic context.

- ## SDG Alignment

This research aligns directly with United Nations Sustainable Development Goal (SDG) 4: Quality Education, which promotes inclusive, equitable, and quality education and encourages the use of innovative technology to improve learning outcomes and broaden access to educational resources. By reducing information overload and enabling students to study more efficiently using AI-generated, objective-aligned materials, the proposed system contributes to improving the quality of higher education learning experiences, particularly for students in developing nations such as Sri Lanka where access to personalized academic support tools remains limited.

- ## Research Cluster Alignment

This research falls within the Intelligent Systems and Educational Technology research cluster, which encompasses the application of AI, NLP, and machine learning techniques to solve real-world problems in digital learning environments. The proposed work specifically contributes to the sub-areas of intelligent tutoring systems, automated content generation, and learning analytics, where the goal is to develop smart, context-aware tools that enhance student engagement, academic performance, and knowledge retention in higher education settings.

1.2 Literature Review

Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) have significantly influenced modern digital education by enabling automated content generation, intelligent tutoring, and personalized learning environments. Recent research explores the integration of AI into learning management systems (LMS) and educational platforms to improve students' access to learning resources and enhance learning efficiency.

Several studies focus on the application of NLP technologies in education to support automated processing of academic content. Research on NLP applications in academic environments shows that NLP techniques can analyze educational materials, extract key concepts, and assist students in understanding complex information more efficiently. These systems help improve academic efficiency by enabling automated summarization, information retrieval, and learning assistance tools for students. However, the research highlights that many NLP-based tools still generate generalized outputs without considering specific academic learning objectives or assessment priorities[1]

Similarly, AI-powered student learning systems have been developed to support adaptive and personalized learning. These systems use machine learning models to analyze learning materials and student interactions to provide customized learning experiences. Such platforms can recommend relevant study materials and generate interactive learning resources that support student engagement. Despite these capabilities, existing student learning systems primarily focus on adaptive learning recommendations rather than generating structured summaries aligned with course learning outcomes[2]

Another research direction explores the integration of AI and NLP within e-learning platforms. These platforms use AI technologies to automate content delivery, generate learning materials, and improve the accessibility of online learning environments. Studies show that AI-enabled e-learning platforms can support automated tutoring, content summarization, and intelligent feedback systems. However, many existing platforms still rely on general-purpose content generation methods that do not incorporate curriculum-level learning objectives when producing study materials[3]

Recent studies also highlight the growing impact of generative AI in education, where large language models are used to automatically generate explanations, summaries, and practice questions from learning materials. Generative AI technologies have the potential to transform digital learning by providing real-time academic assistance and scalable learning resources. However, researchers emphasize the need for better alignment between AI-generated educational content and pedagogical goals to ensure that generated materials support intended learning outcomes[4]

Several systems have been developed as AI-powered study companions that assist students with reviewing academic materials. For example, AI-based study companion systems analyze course content and generate summaries, flashcards, and explanations to support student revision. These systems demonstrate the potential of NLP algorithms to improve study efficiency and knowledge retention. Nevertheless, most existing study companion tools focus on general summarization and content assistance rather than emphasizing exam-oriented learning objectives[5]

Another important research approach is the development of AI-driven intelligent tutoring systems. These systems provide personalized support by analyzing learning materials and interacting with students using conversational interfaces. Intelligent tutoring systems can adapt explanations to learner needs and improve understanding of complex subjects. Research also shows that AI tutors can enhance the teaching–learning process by providing continuous feedback and interactive learning experiences in digital environments[6]

More advanced research explores retrieval-augmented generation (RAG)–based intelligent tutoring systems, where AI models retrieve relevant information from educational resources and generate contextual explanations or answers. These systems improve the accuracy and relevance of AI responses by combining information retrieval with language generation models. Although such systems provide more reliable educational assistance, they mainly focus on tutoring and concept explanation rather than generating structured revision materials aligned with course outcomes[7]

Additionally, AI-powered learning management systems with generative capabilities have been proposed to dynamically generate course materials such as quizzes, flashcards, and summaries. These platforms integrate AI chatbots and automated testing modules to support students in reviewing course content and evaluating their knowledge. Research shows that such systems can significantly enhance student engagement and interactive learning experiences within digital education environments[8]

Furthermore, LLM-driven educational chatbots have been developed to assist students with course-related questions and explanations. For instance, Moodle Bot demonstrates how conversational AI can provide real-time academic support by answering queries, generating exercises, and guiding students through course materials. Although these systems effectively support learning interactions, they mainly function as tutoring assistants and do not specifically generate summaries or flashcards aligned with learning objectives[9]

1.3 Research Gap

- **Research Value**

Recent studies demonstrate that Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) are increasingly applied in digital education systems to improve learning accessibility and efficiency. NLP techniques have been used to analyze academic documents and extract key information from learning materials, enabling automated academic support tools for students [1]. AI-powered student learning systems and adaptive platforms personalize learning experiences by analyzing student interactions and recommending relevant educational resources [2].

AI integration in e-learning environments enables automated content delivery, interactive learning resources, and scalable educational support [3]. Generative AI further allows automatic generation of explanations, summaries, and practice questions from educational materials [4], while AI-powered study companions produce summaries and flashcards to enhance revision efficiency [5]. Advanced systems such as intelligent tutoring systems and retrieval-augmented generation (RAG) models provide contextual explanations and personalized learning support [7]. Learning management systems and LLM-driven chatbots also offer automated quizzes, summaries, and real-time assistance to students [8][9], and intelligent tutors adapt explanations to individual learning needs [6].

However, the scope of these studies is primarily limited to general content generation, conversational tutoring, or adaptive learning support. Existing systems do not explicitly incorporate course learning objectives or curriculum priorities when generating study materials. Consequently, AI-generated summaries often lack focus on examinable concepts and may include irrelevant information, reducing their usefulness for assessment-oriented revision.

Within the scope of this research, the focus is on lecture-based university courses where learning objectives are explicitly defined in syllabi. The proposed system aims to bridge the gap between AI-generated content and outcome-based education by integrating learning objectives into the summarization process. By prioritizing content aligned with course outcomes, the research introduces a novel approach that transforms AI summarization from generic information compression into targeted academic support for examination preparation.

.

- **Product Value**

From a practical perspective, current AI educational products mainly function as tutoring assistants, content generators, or adaptive learning systems. Although they can produce summaries, quizzes, and flashcards, they rarely support outcome-based revision where students must focus on concepts directly linked to course objectives and assessments.

In real academic contexts, especially in higher education, students often face large volumes of lecture materials and limited time for revision. Existing AI tools may generate extensive outputs but lack mechanisms to prioritize content according to instructional goals. This limits their effectiveness as practical study tools.

The scope of the proposed product focuses on university-level lecture materials such as slides, transcripts, and course notes. It is designed primarily for exam preparation and revision rather than full course delivery or tutoring. By aligning generated summaries and flashcards with learning objectives, the system provides structured, exam-focused study resources that directly support curriculum outcomes.

This targeted scope enables the system to deliver clear practical value. Students can quickly identify key concepts required for assessments, reducing cognitive overload and improving revision efficiency. Additionally, the system can be integrated into learning management systems or deployed as a standalone study assistant, making it suitable for both institutional and individual use.

2.1 OBJECTIVE

2.1 Main Objective

To develop and evaluate an AI-based learning-objective–aware lecture summarization and flashcard generation system that analyzes lecture materials and automatically generates ranked summaries and exam-oriented flashcards aligned with course learning objectives to improve students' revision efficiency and learning outcomes.

2.2 Specific Objectives

1. **To collect and preprocess lecture materials**, including lecture slides, transcripts, and course learning objectives, to prepare structured input data for the summarization system.

2. **To design and implement a Natural Language Processing (NLP) model** that extracts key concepts and important information from lecture materials.

3. **To develop a learning-objective alignment mechanism** that identifies and prioritizes lecture content related to course learning outcomes.

4. **To generate ranked lecture summaries** that highlight the most important concepts based on their relevance to learning objectives.

5. **To automatically generate exam-oriented flashcards** from the summarized lecture content to support efficient student revision.

6. **To evaluate the effectiveness of the proposed system** using quizzes and student feedback to measure improvements in learning efficiency and knowledge retention.

3. METHODOLOGY

3.1 Research Approach

This research will follow a Design Science Research (DSR) approach, which focuses on designing, developing, and evaluating an innovative technological artifact to solve a specific problem. The artifact in this study is an AI-based lecture summarization and flashcard generation system that aligns generated study materials with course learning objectives. The design science approach is appropriate because the research aims to create and evaluate a functional system that improves students' learning and revision efficiency.

The research process will include the following stages:

1. **Problem Identification** – Identifying the limitations of existing AI-based summarization systems in educational contexts.
2. **System Design** – Designing a system architecture that integrates lecture materials with course learning objectives.
3. **Implementation** – Developing the AI-based summarization and flashcard generation modules.
4. **Evaluation** – Assessing the effectiveness of the system through quizzes and student feedback.

3.2 Technical Methods and Algorithms

The proposed system will use Natural Language Processing (NLP) and Large Language Model (LLM)–based text processing techniques to analyze lecture materials and generate summaries and flashcards.

The main technical steps include:

3.2.1 Data Preprocessing

- Extract text from lecture slides and transcripts.
- Clean and normalize text using NLP preprocessing techniques such as tokenization, stop-word removal, and sentence segmentation.

3.2.2 Lecture Summarization

- An AI-based summarization model will generate concise summaries by prioritizing content that aligns with learning objectives.
- Ranked summaries will highlight the most important exam-relevant concepts.

3.2.3 Learning Objective Alignment

- Semantic similarity techniques will be used to match lecture content with course learning objectives.
- Embedding models will compare lecture sentences with learning objectives to determine relevance.

.

3.2.4 Flashcard Generation

- The summarized content will be transformed into question–answer flashcards using generative AI techniques.
- Flashcards will focus on key definitions, concepts, and explanations.

3.3 Tools and Platforms

The system will be implemented using modern AI and web development technologies.

Programming Language

- Python – widely used for NLP and machine learning applications.

NLP and AI Libraries

- SpaCy or NLTK – for text preprocessing and NLP tasks.
- Transformers / Large Language Models – for summarization and flashcard generation.

Development Framework

- Flask or FastAPI – for building the backend system and APIs.

Data Processing Tools

- PDF text extraction libraries for processing lecture slides.

These tools are selected because they provide reliable NLP capabilities and are widely used in AI-based research and development.

3.4 Data Collection and Ethical Considerations

The data used in this research will consist of lecture materials collected from university courses, including lecture slides, transcripts, and course learning objectives.

To ensure ethical research practices:

- Only non-sensitive educational materials will be used.
- Any personal student information will not be collected or stored.
- Participants involved in evaluation (students) will provide voluntary consent before participating in quizzes or surveys.
- All collected data will be used solely for academic research purposes.

3.5 Validation Strategy and Performance Metrics

The proposed system will be evaluated using both quantitative and qualitative evaluation methods.

1. Quiz-Based Evaluation

Students will study lecture materials using the generated summaries and flashcards. Their performance will be measured through quizzes to evaluate knowledge retention.

2. Student Feedback

A survey will be conducted to collect student opinions regarding the usefulness, clarity, and effectiveness of the generated summaries and flashcards.

3. Performance Metrics

The system will be evaluated using the following metrics:

- **Summary Relevance Score** – Measures how well generated summaries align with learning objectives.
- **Quiz Performance Improvement** – Compares student performance before and after using the system.
- **User Satisfaction Score** – Based on student feedback surveys.
- **Flashcard Accuracy** – Measures correctness and relevance of generated flashcards.

These evaluation methods will help determine whether the proposed system improves students' ability to review lecture materials efficiently and prepare for examinations.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

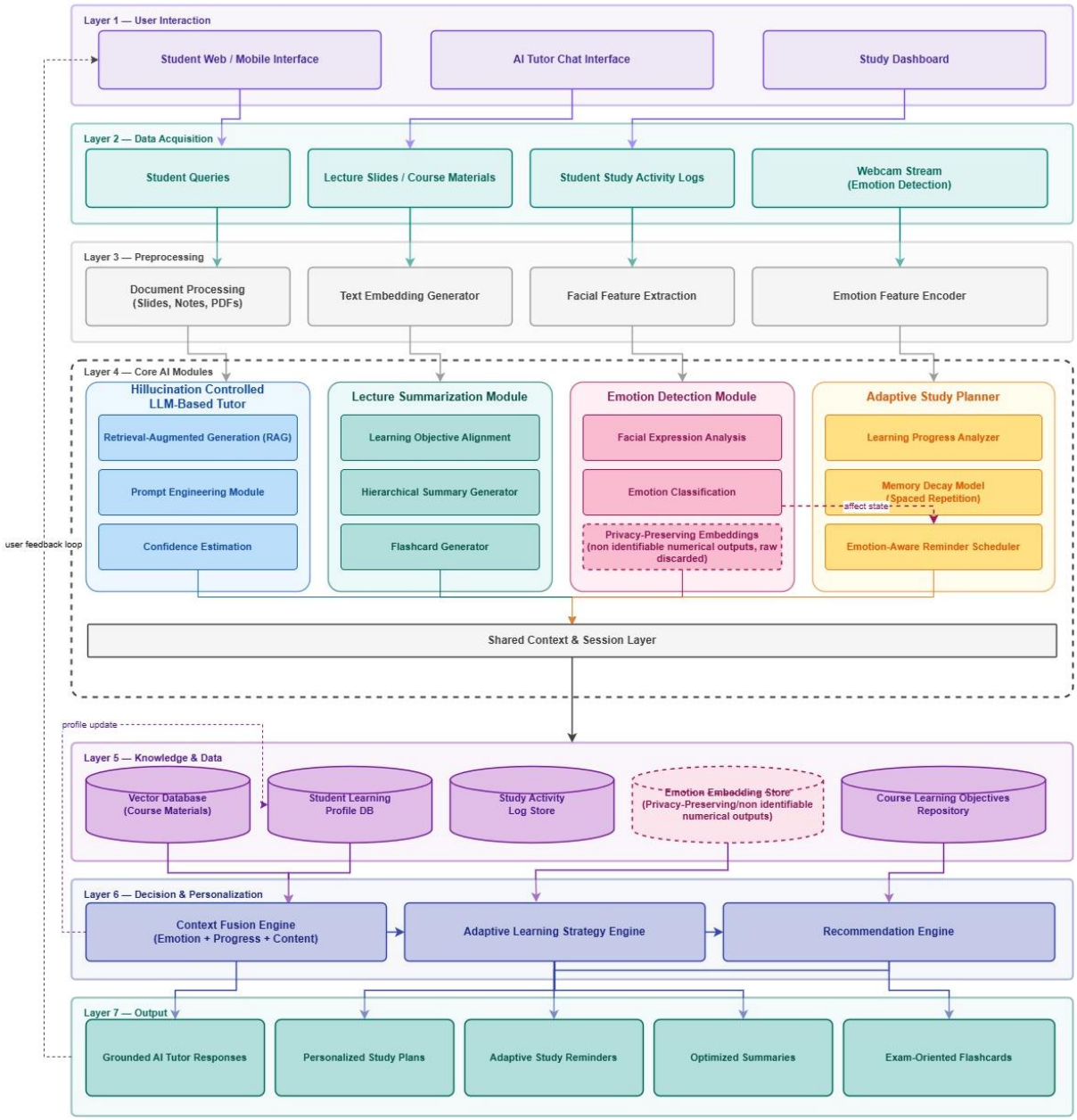


Figure 1. Overall System Architecture – Smart Study Assistant (R26-SE-022)

4.1 Overview of Proposed Solution

The proposed system is designed to transform large volumes of lecture materials into structured revision resources using Artificial Intelligence and Natural Language Processing techniques. The system processes lecture slides, transcripts, and related academic documents to automatically generate concise summaries and exam-oriented flashcards that support efficient learning and revision.

The architecture of the proposed system consists of several interconnected modules including document processing, text preprocessing, summarization, learning objective alignment, and flashcard generation. These modules work together to extract meaningful information from lecture materials and convert them into structured study resources that students can easily review.

The system begins by accepting lecture materials such as lecture slides and transcripts as input. These materials are first processed to extract textual content. After extraction, the text undergoes Natural Language Processing (NLP) preprocessing techniques including tokenization, cleaning, and sentence segmentation to prepare the data for further analysis. The processed text is then passed to the summarization engine where a transformer-based model generates concise summaries of the lecture content.

To ensure that the summaries focus on exam-relevant information, the system integrates course learning objectives into the summarization process. This allows the system to prioritize concepts that align with academic outcomes. Finally, the summarized concepts are converted into flashcards that help students review important topics using active recall learning techniques.

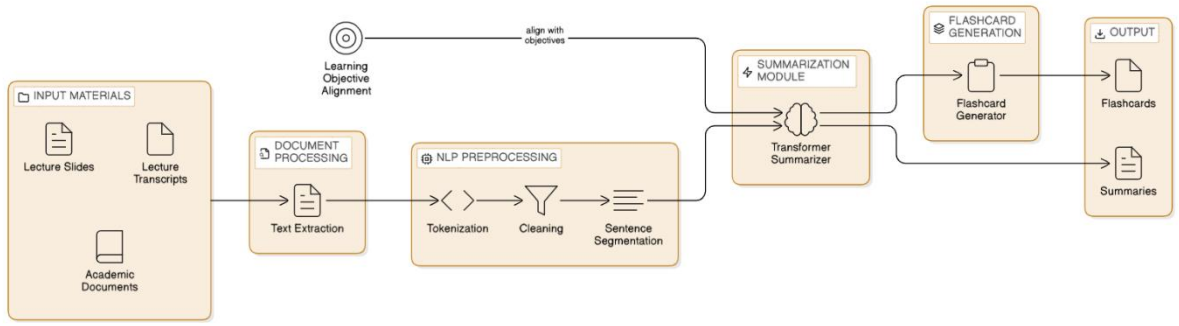


Figure 4. 1 **High-Level Architecture of the Proposed Lecture Summarization System**

The architecture illustrated in Figure 4.1 demonstrates the overall workflow of the proposed system. Lecture materials are first processed through document extraction and NLP preprocessing stages before being passed to the summarization module. The learning objective alignment component ensures that summaries remain relevant to course learning outcomes.

Finally, the flashcard generator produces structured study materials that can be accessed through a student interface.

4.2 Component Description

The proposed system consists of several functional components that work together to generate structured learning materials from lecture documents.

The **document processing module** is responsible for extracting textual information from lecture slides, PDFs, and transcripts. This module converts various document formats into machine-readable text suitable for Natural Language Processing tasks.

The **text preprocessing module** performs standard NLP preprocessing operations such as tokenization, stop-word removal, sentence segmentation, and text normalization. These processes clean the extracted text and prepare it for the summarization stage.

The **summarization engine** is the core component of the system. This module uses transformer-based models such as BART or T5 to generate concise summaries from lecture content. These models analyze contextual relationships within the text and identify the most relevant sentences that represent the key concepts of the lecture.

The **learning objective alignment module** ensures that generated summaries emphasize exam-relevant topics. This component compares summarized concepts with predefined course learning objectives using semantic similarity techniques. As a result, the system prioritizes concepts that are directly aligned with expected learning outcomes.

The **flashcard generation module** converts summarized concepts into question-answer pairs. These flashcards allow students to quickly review important concepts and improve their knowledge retention through active recall learning techniques.

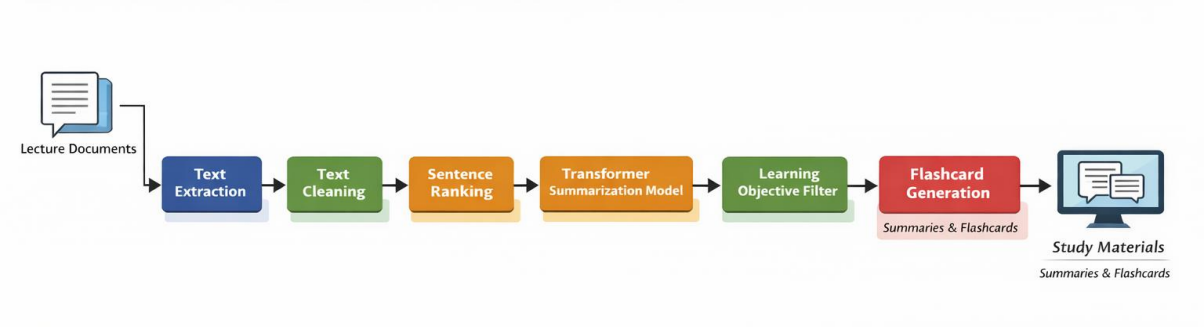


Figure 4. 2 **Internal Architecture of the Lecture Summarization and Flashcard Generation Module**

Figure 4.2 illustrates the internal workflow of the lecture summarization module. Lecture documents are first processed through text extraction and cleaning stages before important

sentences are identified. The transformer summarization model generates concise summaries, which are then filtered using learning objectives. Finally, the flashcard generator produces structured study materials that students can use for revision.

4.3 Component Interactions

The components of the proposed system interact through a sequential data processing pipeline. The document processing module extracts lecture text and sends it to the preprocessing module. After preprocessing, the cleaned text is forwarded to the summarization engine. The summarization module generates concise summaries which are then evaluated by the learning objective alignment module to ensure relevance to course learning outcomes. Finally, the flashcard generation module converts important summarized concepts into question-answer flashcards that can be presented to students through the system interface.

4.4 Data Flow Explanation

The data flow of the proposed system begins with lecture materials being uploaded into the system. The document processing module extracts textual information and sends it to the NLP preprocessing module. After preprocessing, the text is analyzed by the summarization engine to generate concise summaries. These summaries are then aligned with course learning objectives to prioritize important topics. The resulting concepts are used to generate flashcards which are provided to students as structured study materials.

4.5 Scalability and Security Considerations

The proposed system is designed to support scalability by utilizing modular architecture and cloud-based deployment options. Transformer-based models can be deployed using scalable machine learning frameworks that allow the system to process large volumes of lecture materials efficiently.

Security considerations are also incorporated into the system design. Lecture documents will be stored securely, and access to the system will be controlled through authentication mechanisms. Additionally, the system will avoid storing sensitive personal information and will comply with ethical data handling practices when processing academic materials.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed lecture summarization and flashcard generation system involves several stakeholders who will interact with or benefit from the system. These stakeholders are categorized as primary and secondary users based on their level of interaction with the system.

Primary Stakeholders

• **University Students** – The main users of the system. Students will use the platform to upload lecture materials, generate summaries, and access flashcards for exam preparation. The system aims to help students reduce information overload and improve their revision efficiency.

Secondary Stakeholders

• **Lecturers / Instructors** – Lecturers may use the system to assist students in reviewing lecture materials and to ensure that the generated summaries align with course learning objectives.

• **Educational Institutions** – Universities and educational organizations may integrate the system into their learning management platforms to enhance digital learning support for students.

• **EdTech Developers** – Educational technology companies may adopt or extend the system to develop intelligent study assistance tools.

5.2 Functional Requirements

Functional requirements describe the specific capabilities and functions that the system must perform.

FR1: The system shall allow users to upload lecture materials such as lecture slides, PDFs, or transcripts.

FR2: The system shall extract textual information from uploaded lecture documents.

FR3: The system shall preprocess extracted text using Natural Language Processing techniques including tokenization and text cleaning.

FR4: The system shall generate concise summaries from lecture materials using a transformer-based summarization model.

FR5: The system shall align generated summaries with predefined course learning objectives.

FR6: The system shall identify key concepts from summarized lecture content.

FR7: The system shall automatically generate flashcards in question-and-answer format based on identified key concepts.

FR8: The system shall present summaries and flashcards through a user interface for student revision.

FR9: The system shall allow users to review generated flashcards for active recall learning.

5.3 Non-Functional Requirements

Non-functional requirements describe system quality attributes such as performance, security, usability, and reliability.

Performance Requirements

NFR1: The system shall generate lecture summaries within a reasonable processing time (less than 10 seconds for standard lecture documents).

NFR2: The system shall handle large lecture documents efficiently without significant performance degradation.

Security Requirements

NFR3: The system shall ensure secure storage of uploaded lecture materials.

NFR4: The system shall prevent unauthorized access to uploaded documents and generated summaries.

Usability Requirements

NFR5: The system interface shall be user-friendly and easy for students to operate without technical knowledge.

NFR6: The system shall present summaries and flashcards in a clear and readable format to support effective revision.

Scalability Requirements

NFR7: The system shall support scalability to accommodate multiple users simultaneously.

Reliability Requirements

NFR8: The system shall provide consistent and accurate summary outputs during repeated operations.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed lecture summarization and flashcard generation system includes university students and higher education institutions that require efficient learning support tools.

The main customer persona is university students who struggle to review large amounts of lecture materials before examinations. These students often rely on lecture slides, notes, and recorded lectures to prepare for assessments, which can be time-consuming and overwhelming.

Secondary target users include online learning platforms, educational institutions, and EdTech companies that aim to provide AI-based learning assistance tools to students. These organizations can integrate the proposed system into their learning management systems to improve student engagement and academic performance.

6.2 Value Proposition

The proposed system provides a unique value proposition by automatically transforming lecture materials into structured revision resources. Unlike traditional summarization tools, the system integrates course learning objectives to ensure that summaries focus on exam-relevant topics.

Additionally, the system generates flashcards from summarized concepts, enabling students to engage in active recall learning, which improves knowledge retention and learning efficiency. This combination of AI-driven summarization and automated flashcard generation provides students with a powerful revision tool that reduces information overload and improves study effectiveness.

6.3 Revenue Model

The proposed system can adopt several revenue models depending on the target market and deployment environment.

One possible revenue model is a subscription-based service, where students pay a monthly or yearly fee to access premium features such as advanced summarization, personalized flashcards, and integration with academic learning platforms.

Another potential revenue model is institutional licensing, where universities or educational institutions purchase licenses to integrate the system into their learning management systems.

The system may also generate revenue through freemium services, where basic features are provided for free while advanced features require paid subscriptions.

6.4 Cost Estimation

The development and deployment of the proposed system require several resources including computing infrastructure, software tools, and maintenance support.

Major costs may include:

- Cloud computing resources for AI model processing
- Software development tools and libraries
- Data storage and hosting infrastructure
- Maintenance and system updates

These costs are expected to be moderate since many Natural Language Processing libraries and AI frameworks used in the project are open-source.

6.5 Pricing Strategy

The proposed system can follow a tiered pricing strategy to attract a broad range of users.

For example:

• Free basic version with limited summarization capabilities
• Student subscription plan with advanced summarization and flashcard generation
• Institutional packages for universities and educational platforms

Affordable pricing is important to ensure accessibility for students while still supporting system sustainability.

6.6 Competitive Advantage

The proposed system has several competitive advantages compared with existing AI-based study tools.

First, it introduces a learning-objective-aware summarization mechanism, which ensures that generated summaries focus on exam-relevant content rather than general text compression.

Second, the system integrates automatic flashcard generation, allowing students to review key concepts through active recall learning.

Third, the system is designed specifically for academic lecture materials, making it more suitable for educational environments than general-purpose summarization tools.

These innovations allow the proposed system to provide more targeted and effective revision support for students.

6.7 Intellectual Property Considerations

The proposed system may involve intellectual property considerations related to the AI algorithms and software implementation used for lecture summarization and flashcard generation.

If the system demonstrates significant innovation, the software architecture and algorithms may be eligible for software copyright protection or patent consideration. Additionally, the system can be distributed under appropriate software licensing agreements when deployed commercially.

Proper licensing and ethical data usage practices will also be maintained to ensure compliance with intellectual property laws and academic research standards.

7. BUDGET AND JUSTIFICATION

This section outlines the estimated costs required to design, develop, test, and evaluate the Learning-Objective-Aware Lecture Summarization and Flashcard Generation System. The budget includes hardware, software tools, cloud infrastructure, and human development effort required for implementing the AI-based summarization system.

1 Hardware Costs

Item	Estimated Cost (LKR)	Justification
Development Laptop / Workstation	150,000 – 200,000	Required for implementing NLP models, preprocessing lecture materials, and backend development. A GPU-enabled machine helps speed up text processing and model experimentation.
External Storage (SSD)	15,000 – 25,000	Used to store lecture slides, transcripts, embeddings, generated summaries, and experimental datasets.

Justification

Hardware resources are required to perform local development, text preprocessing, and AI model testing before deploying the system to cloud infrastructure. A high-performance workstation enables efficient processing of lecture materials and improves development speed.

2 Software Licenses

Software / Tool	Estimated Cost (LKR)	Justification
Hugging Face Model Access	Free	Provides access to pre-trained NLP models used for summarization and flashcard generation.
Vector Database (FAISS / ChromaDB)	Free	Open-source tools used to store embeddings and manage lecture content efficiently.
API Testing Tools (Postman)	Free	Used for testing system APIs and ensuring proper communication between components.

Justification

Most modern AI development tools are open source, which significantly reduces development costs. Optional premium access may be required for advanced models or higher API usage during experimentation.

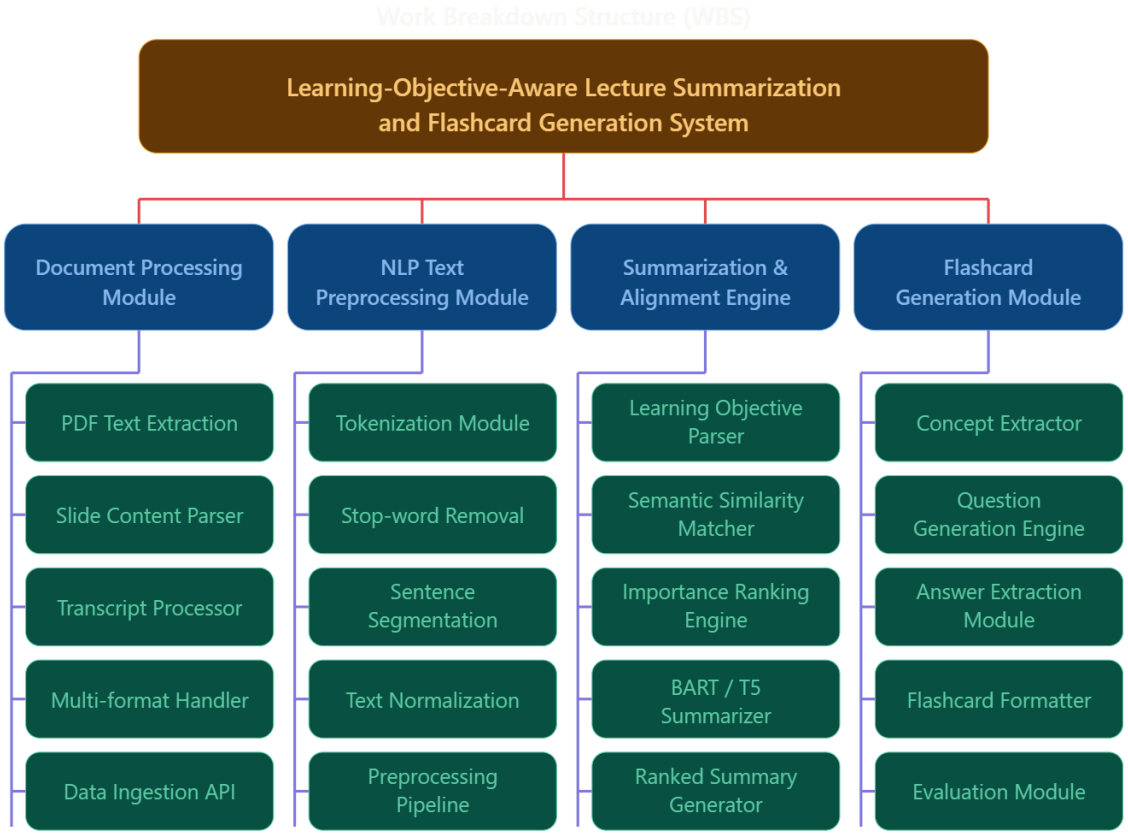
3 Cloud Services

Service	Estimated Cost (LKR)	Justification
Cloud Compute (AWS / Azure / GCP)	40,000 – 70,000/ month	Used to host the backend system, AI summarization models, and APIs.
Storage (S3 / Blob Storage	5,000 – 10,000 / month	Used to store lecture slides, transcripts, generated summaries, and system logs.
Model Inference Costs	10,000 – 20,000 / month	Required for running AI summarization models and generating flashcards dynamically.

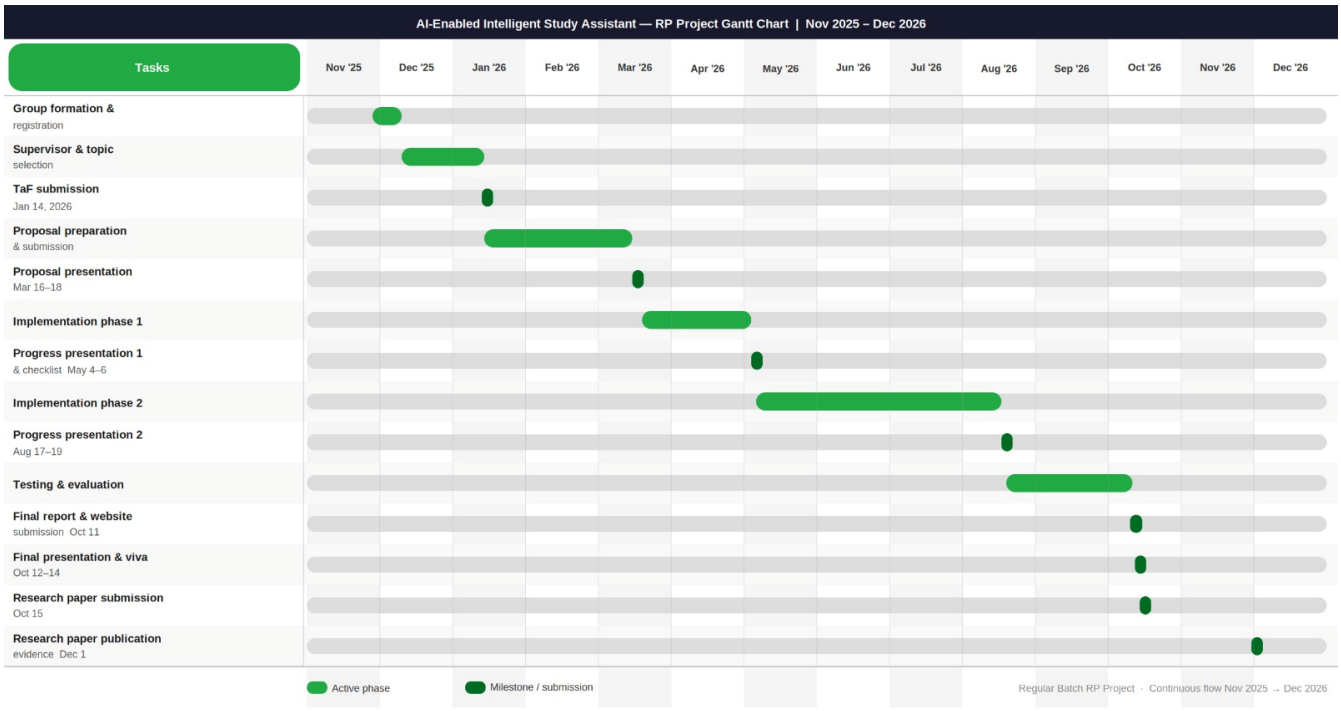
Justification

Cloud services provide scalability, reliability, and availability. They allow the system to process multiple lecture materials, generate summaries for multiple users, and store academic content securely.

8.WORK BREAKDOWN STRUCTURE (WBS)



9.GANTT CHART



10. References List

[1] A. Singh, A. Singh, and H. Kumar, "Natural Language Processing and its Impact on Academics and on Student's Efficiency and Effectiveness," in *2025 International Conference on Circuit, Systems and Communication (ICCSC)*, Fez, Morocco: IEEE, Jun. 2025, pp. 1–5. doi: 10.1109/ICCSC66714.2025.11135378.

[2] A. Sabar, S. Gamage, S. Abishek, C. Lakmal, K. A. D. T. Kulawansa, and P. Perera, "AI-Powered Student Learning System," in *2024 8th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI)*, Ratmalana, Sri Lanka: IEEE, Dec. 2024, pp. 1–5. doi: 10.1109/SLAAI-ICAI63667.2024.10844979.

[3] M. Kaouni, F. Lakrami, and O. Labouidya, "Integrating Artificial Intelligence and Natural Language Processing in E-Learning Platforms: A Review of Opportunities and Limitations," in *2023 7th IEEE Congress on Information Science and Technology (CiSt)*, Agadir - Essaouira, Morocco: IEEE, Dec. 2023, pp. 56–63. doi: 10.1109/CiSt56084.2023.10409971.

[4] M. Sindhu, Sarita, Y. Y. Raghav, and K. Kundu, "Generative AI in Education and its Future Possibilities," in *2025 International Conference on Intelligent and Secure Engineering Solutions (CISES)*, Greater Noida Gautam Budh Nagar, India: IEEE, Aug. 2025, pp. 68–74. doi: 10.1109/CISES66934.2025.11265181.

[5] A. Aher, R. Sawant, R. Patil, S. Sabat, and T. Panchal, "Mindscape: An AI-Powered Study Companion Integrating Natural Language Processing Algorithms," in *2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI)*, Greater Noida, India: IEEE, Apr. 2025, pp. 2116–2121. doi: 10.1109/ICCSAI64074.2025.11063938.

[6] V.-P. Sorescu and D. Andone, "Developing Intelligent Tutors with Artificial Intelligence for Digital Education," in *2025 IEEE Digital Education and MOOCS Conference (DEMOcon)*, Boca Raton, FL, USA: IEEE, Oct. 2025, pp. 1–6. doi: 10.1109/DEMOcon65705.2025.11282632.

[7] Y. Obata and M. Hamada, "Adaptive Retrieval-Augmented Generative Intelligent Tutoring System for Automata Theory," in *2025 IEEE 18th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC)*, Singapore, Singapore: IEEE, Dec. 2025, pp. 477–482. doi: 10.1109/MCSoC67473.2025.00080.

[8] V. K, M. T. N, S. Nithyashri, A. S. Gopal, and S. R P, "Dynamic Content Generation for Learning Management System Using Generative AI," in *2025 International Conference on Computing for Sustainability and Intelligent Future (COMP-SIF)*, Bangalore, India: IEEE, Mar. 2025, pp. 1–7. doi: 10.1109/COMP-SIF65618.2025.10969929.

[9] A. T. Neumann, Y. Yin, S. Sowe, S. Decker, and M. Jarke, "An LLM-Driven Chatbot in Higher Education for Databases and Information Systems," *IEEE Trans. Educ.*, vol. 68, no. 1, pp. 103–116, Feb. 2025, doi: 10.1109/TE.2024.3467912.

9. APPENDICES

- This section includes additional supporting materials related to the proposed research project, including survey questions, system diagrams, preliminary user interface designs, and risk analysis.

Appendix A – Survey Instrument

- A short survey will be conducted to evaluate the usefulness of the proposed lecture summarization system. Example questions include:
  1. Do you find it difficult to review large lecture materials before exams?
  2. Would AI-generated summaries help you revise faster?
  3. How useful would flashcards be for revising lecture concepts?

Appendix B – System Diagrams

- This appendix includes additional system diagrams that illustrate the internal architecture and data flow of the proposed system.

- The proposed system follows a pipeline-based architecture consisting of five interconnected modules. The Document Processing Module accepts lecture slides (PDF/PPTX) and transcripts as inputs and extracts raw textual content. This extracted text is passed to the NLP Preprocessing Module, which performs tokenization, stop-word removal, and sentence segmentation to clean and normalize the data. The cleaned text is then processed by the Summarization Engine, which uses transformer-based models such as BART or T5 to generate concise candidate summaries. These summaries are filtered and prioritized by the Learning Objective Alignment Module, which uses semantic similarity techniques to map lecture content to predefined course learning objectives. Finally, the Flashcard Generation Module converts the ranked, objective-aligned summaries into question-and-answer pairs that are presented to the student through the user interface.

- The data flow follows a strictly sequential pipeline: uploaded materials are extracted, cleaned, summarized, aligned with learning objectives, and converted into flashcards before being delivered to the student for revision.

Appendix C – Preliminary UI Mockups

- This section presents simple interface designs for the proposed system. The UI is designed to be intuitive and accessible for university students without requiring technical knowledge.

- **Screen 1 – Lecture Material Upload Page:** This screen allows students to upload lecture materials for processing. It includes a file upload area supporting PDF, PPTX, and TXT formats, a course title and module name input field, an optional learning objective input field for alignment, and a "Generate Summary" action button with an upload progress indicator.

- **Screen 2 – Generated Summary Page:** This screen displays the AI-generated, objective-aligned ranked summary to the student. It features a summary panel showing concise lecture concepts ranked by relevance, importance badges (High / Medium / Low) assigned to each concept, learning objective tags displayed alongside matched content, a download or save option, and a "Generate Flashcards" button to proceed to the next step.

- **Screen 3 – Flashcard Review Interface:** This screen allows students to review flashcards using active recall learning techniques. It includes a flashcard flip mechanism where the question is displayed on the front and the answer is revealed on flip, Previous and Next navigation controls, a progress indicator showing the current card number, self-rating buttons (Knew it / Unsure / Didn't know) to support spaced repetition, and an option to export flashcards in PDF or CSV format.

Appendix D – Risk Analysis

Risk	Mitigation
• Poor lecture data quality	• Use preprocessing and filtering
• Model accuracy issues	• Evaluate using ROUGE metrics
• Low user adoption	• Conduct user feedback testing

Appendix E – Ethical Considerations

- The research will use only academic lecture materials and will not collect personal student data. Any survey responses collected will remain anonymous and used only for research purposes.
- All student participants involved in system evaluation will be fully informed about the purpose of the research prior to participation. Participation will be entirely voluntary, and participants may withdraw at any time without penalty. A formal consent form will be obtained from each participant before any data collection activity takes place.

- The AI models used in this research will be applied solely for educational summarization and flashcard generation. The accuracy and relevance of all AI-generated outputs will be validated through manual human evaluation to ensure reliability and prevent the production of misleading academic content. All lecture materials used will be obtained with appropriate institutional permission or accessed through publicly available licensed academic sources, ensuring full compliance with copyright obligations.
- This research aligns with the SLIIT Research Ethics Policy and supports UN Sustainable Development Goal 4 (Quality Education) by improving equitable access to effective AI-assisted academic study tools.
-